CH1 SIGNAL PATH (replicate x4 for CH1–CH4)

STAGE 1: INA128
In-Amp G=51
RG=1kΩ 1% MF
(pins 1–8)

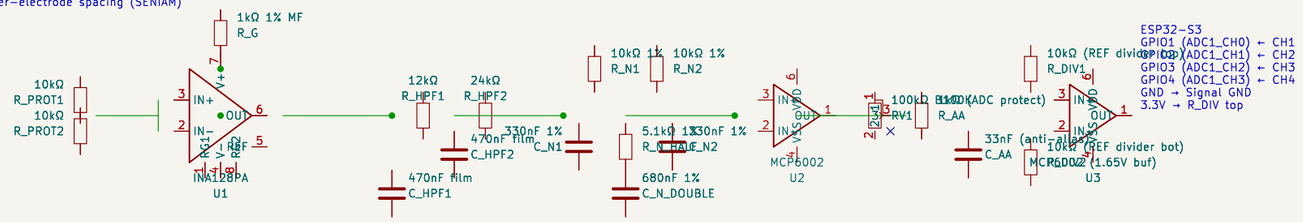
STAGE 2: Sallen-Key
HPF fc=20Hz
C=470nF (film)
R1=12kΩ, R2=24kΩ

STAGE 3: Twin-T
Notch fc=50Hz
R=10kΩ 1%, R/2=5.1kΩ
C=330nF, 2C=680nF

STAGE 4: Var Gain
MCP6002 x1–x10
RV=100kΩ (B100K)
Anti-alias LPF 100Hz

STAGE 5: ADC
Level-shift 1.65V
Series 1kΩ + 33nF
ESP32-S3 ADC1

ELECTRODE INPUT
(Ag/AgCl differential pair)
20mm inter-electrode spacing (SENIAM)



NOTE: Repeat U1–RV1 block for CH2 (GPIO2), CH3 (GPIO3), CH4 (GPIO4)
U3 (1.65V buffer) is SHARED across all 4 channels
All INA128 REF pins connect to U3 output (1.65V)
Supply: 9V battery → LM7805 → 5V → U1,U2,U3 VCC
Use separate analog GND plane; join to digital GND at single star point

PCA9685 SERVO DRIVER (shared)
VCC=5V, GND, SDA→GPIO8, SCL→GPIO9
CH0–CH4: SG90 fingers (0°–180°)
CH5: MG996R wrist
POWER: 5V 5A+

COMPONENT REFERENCE SUMMARY
R_G: 1kΩ 1% MF (G=51 for INA128) × 4
R_PROT: 10kΩ × 8 (input protection, 2 per channel)
C_HPF: 470nF film × 8 (Sallen-Key HPF, 2 per channel)
R_HPF1: 12kΩ × 4; R_HPF2: 24kΩ × 4 (fc=20Hz)
R_N1,R_N2: 10kΩ 1% × 8; R_N_HALF: 5.1kΩ 1% × 4 (Twin-T)
C_N1,C_N2: 330nF 1% × 8; C_N_DOUBLE: 680nF 1% × 4 (Twin-T)
RV1: 100kΩ B100K pot × 4 (variable gain)
R_AA: 1kΩ × 4 (ADC series protection)
C_AA: 33nF × 4 (anti-alias cutoff ~145kHz)
R_DIV1,R_DIV2: 10kΩ × 2 (REF = 1.65V divider)
U1: INA128PA DIP-8 × 4; U2: MCP6002 DIP-8 × 4; U3: MCP6002 (REF buffer, shared)

Supply: +9V battery → LM7805 → +5V analog; REF = 1.65V mid-rail from R_DIV
Replicate U1–U5 block x4 for channels CH1–CH4
INA128 G=51 → Sallen-Key BPF (20–100Hz) → Twin-T Notch (50Hz) → Variable Gain → ESP32-S3 AD
Ian Kiarie – Supervisor: Dr. Irene
JKUAT ECE FYP – ENE212-0069/2022

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Title: sEMG 4-Channel Analog Front-End		
Size: A3	Date: 2026-10-18	**Rev: v1.0**
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